Corporate Finance

Beginner-friendly study notes

Built from your three lecture decks: **3A** Present Value & NPV, **3B** Capital Budgeting Rules (IRR, Profitability Index, Payback), **3C** Time-Varying Rates of Return.

How to use these notes

1. Read each part in order. Every part starts with the **plain-English idea** before any formula.
2. Do the **Check yourself** questions at the end of each part. Answers are at the very back.
3. "Units" just means money (the lectures don't name a currency).
4. Don't memorise formulas first. Understand the *why*; the formula is then just shorthand.

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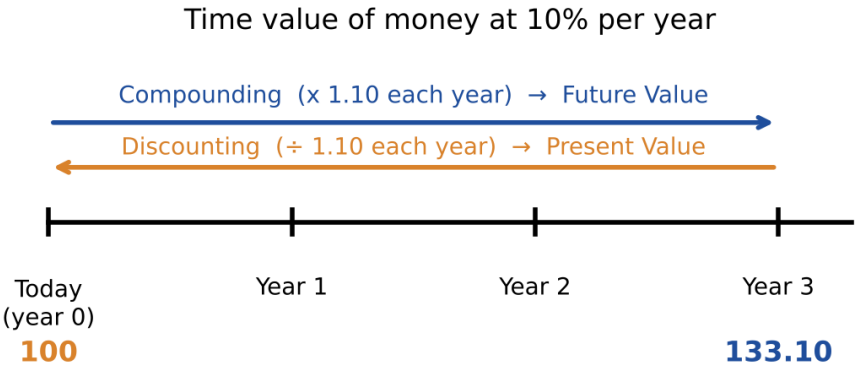
Part	Topic	Lecture
0	The one idea behind everything: money today vs money later	all
1	Present value, annuities, perpetuities, loans, stocks	3A
2	Net Present Value (NPV) and why it is the gold standard	3A
3	Other rules: IRR, Profitability Index, Payback	3B
4	Rates that change over time: compounding & annualizing	3C
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Part 0. The one idea behind everything

Big idea
A rupee (or dollar) today is worth more than a rupee later. Why? Because money today can be invested and grow. So to compare money at different dates, we must convert everything to the same date, usually **today**.

Key words (learn these first)

Word	Meaning in plain English
Interest rate (r)	The reward for waiting. 10% means 100 today becomes 110 next year.
Future Value (FV)	What money today will grow into later.
Present Value (PV)	What money received later is worth **today**.
Discount rate / cost of capital	The rate used to shrink future money back to today. It is the return you could earn elsewhere.
Cash flow (C)	Money coming in (+) or going out (−) at some date. C_0 = today, C_1 = next year.
Compounding	Earning interest on your interest.
Project	Any plan that costs money now and pays back later (a factory, a loan, a stock purchase).



Compounding moves money forward in time; discounting moves it backward.

Worked example: 100 today at 10%
Year 1: 100 × 1.10 = 110. Year 2: 110 × 1.10 = 121. Year 3: 121 × 1.10 = **133.10**.
Going backward: 133.10 received in 3 years is worth 133.10 ÷ 1.10^3 = **100** today.

Formula
$FV = C \times (1 + r)^t$ $PV = C \div (1 + r)^t$ (t = number of years)

Part 1. Present value in action (Lecture 3A)

1.1 Loan repayment (an "annuity")

An **annuity** is a stream of *equal* payments at regular intervals. A home loan is one: you borrow a lump sum today and repay it with equal monthly payments. The amount you borrow is the **present value** of all your future payments.

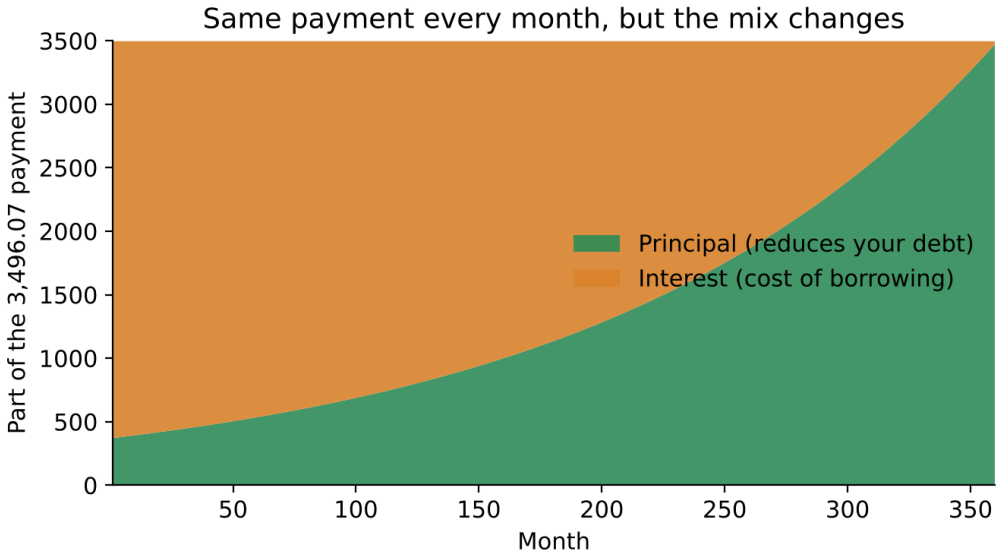
Formula
Loan = $(C \div r) \times [\ 1 - 1 \div (1 + r)^{n}\ ]$ where C = payment, r = rate **per period**, n = number of payments

Worked example: 500,000 loan, 30 years, quoted 7.5% per year (from the lecture)
Banks **quote** the yearly rate by multiplying the monthly rate by 12. So the real monthly rate is 7.5% ÷ 12 = **0.625%**.
n = 30 × 12 = 360 payments. Solve 500,000 = (C ÷ 0.625%) × [1 − 1 ÷ 1.00625^{360}].
Answer: **C ≈ 3,496.07** every month for 360 months.

Interest vs principal: where does each payment go?

- **Interest** = rate × what you still owe (the outstanding balance).
- **Principal** = the rest of the payment; it actually reduces your debt.
- This matters if you repay early, and for tax (interest and principal may be taxed differently).

Month	Balance at start	Interest (0.625%)	Principal repaid	Balance at end
1	500,000.00	3,125.00	371.07	499,628.93
2	499,628.93	3,122.68	373.39	499,255.54
3	499,255.54	3,120.35	375.73	498,879.81



Early payments are mostly interest. Over time, more of each payment is principal.

Big idea
The payment never changes, but as the balance falls, interest falls, so principal rises. That is why paying off a loan in the first years feels slow.

1.2 Perpetuities (payments forever)

A **perpetuity** pays forever. Sounds silly, but the maths is simple and gives quick estimates.

Formula
Constant perpetuity: $PV = C \div r$ Growing perpetuity: $PV = C_1 \div (r - g)$ (needs $r > g$)

C_1 is next year's payment and g is its yearly growth rate.

Worked example: A mature business
Profit next year 1,000,000, growing 2% a year, cost of capital 8%.
Value ≈ 1,000,000 ÷ (8% − 2%) = **16,666,667**. A rough estimate, not an exact value (growth and rates won't stay constant forever).

1.3 Stock valuation: the Gordon growth model

If you believe a stock's dividend grows at g forever, the growing perpetuity formula values the stock. In that setting it has a name: the **Gordon growth model**.

Formula
Price today $P = D \div (r - g)$ D = next year's dividend, r = required return, g = dividend growth

Worked example: Lecture numbers
D = 10, g = 5%, r = 10% → P = 10 ÷ (10% − 5%) = **200**.

Reverse it to find the market's cost of capital

Rearranged: **D ÷ P = r − g**. The left side is the **dividend yield** (next year's dividend ÷ price). So if the yield is 2.2% and dividends are not growing (g = 0%, because the company paid about 6.6 billion both last year and 4 years ago), then r ≈ 2.2% + 0% = **2.2%**. Again an approximation.

Remember
Perpetuity formulas are **back-of-the-envelope tools**. They assume fixed rates and payments forever, which is rarely true.

Check yourself – Part 1

- Q1. What is 500 received in 2 years worth today at 10%?
- Q2. A 200,000 loan over 20 years, quoted 9% per year, monthly payments. Find the payment and the interest part of month 1.
- Q3. A stock pays 50 next year, dividends grow 4% forever, required return 9%. Fair price?

Part 2. Net Present Value (Lecture 3A)

Big idea
NPV = (present value of all money coming in) − (present value of all money going out). It tells you how much richer a project makes you, *in today's money*.

Formula
$NPV = C_0 + C_1 \div (1+r) + C_2 \div (1+r)^2 + C_3 \div (1+r)^3 + \ldots$ (costs are negative numbers)

- **NPV > 0**: take the project.
- **NPV < 0**: reject it.
- **NPV = 0**: you are indifferent.

Worked example: Simple project
Costs 100 now, returns 120 in one year, interest rate 10%.
NPV = −100 + 120 ÷ 1.10 = −100 + 109.09 = **+9.09** → take it.

2.1 Why NPV is a good rule: "free money"

In a perfect market, you can borrow or lend at 6%. Suppose a project costs 1 and pays 1.07 next year. Borrow 1 at 6% (owe 1.06 later), invest it, collect 1.07: you keep **0.01 for free**. A positive-NPV project is free money. A project paying only 1.06 per 1 (zero NPV) gives no free money. And if you can *sell* a project that pays 1.05 for 1, you also pocket 0.01 for free.

- In a perfect world with no uncertainty, positive-NPV projects are rare (everyone would grab them and push returns up until the free money vanishes).
- In the real world NPV is still the key benchmark, but it is hard: you must *estimate* cash flows and discount rates, and markets have taxes, transaction costs, disagreements and market power.

2.2 Separation of decisions

Two people, same project: it costs 100 and returns 200 next year. Each has 150 cash. Interest rate 10%.

	Person A (wants cash today)	Person B (wants to save)
Take project?	Yes. Sell it for 200 ÷ 1.1 = 181.82. Cash today = 181.82 − 100 + 150 = **231.82** (more than 150).	Yes. Next year: 200 + 50 × 1.1 = **255** (more than 150 × 1.1 = 165).
Compare with doing nothing	231.82 > 150	255 > 165

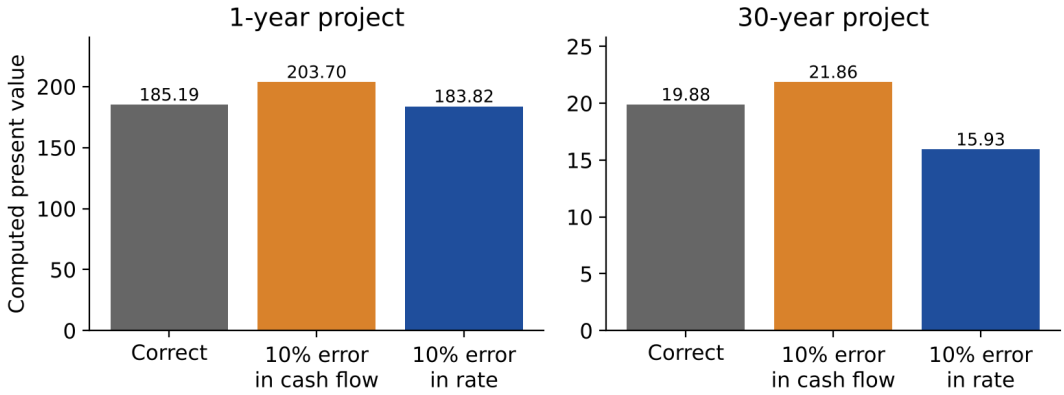
Big idea
Separation of decisions: take every positive-NPV project (investment decision), then use the capital market to shift your money to when you want it (consumption decision). Your personal timing preference should not change which projects you take. (This breaks down in imperfect markets where borrowing and lending rates differ.)

2.3 What if my estimates are wrong? (Sensitivity / scenario analysis)

Scenario (sensitivity) analysis changes one input at a time to see how much the answer moves.

Case	Correct PV	10% error in cash flow (200→220)	10% error in rate (8%→8.8%)
1-year project	200 ÷ 1.08 = 185.19	220 ÷ 1.08 = 203.70 (10% off)	200 ÷ 1.088 = 183.82 (only 1% off)
30-year project	200 ÷ 1.08^{30} = 19.88	220 ÷ 1.08^{30} = 21.86 (10% off)	200 ÷ 1.088^{30} = 15.93 (20% off!)

Which mistake hurts more? (true cash flow 200, true rate 8%)



Over short horizons the cash-flow error dominates; over 30 years the rate error is just as dangerous.

- **Short-term:** getting the interest rate slightly wrong barely matters; getting the cash flow wrong matters a lot.
- **Long-term:** both matter, because the rate is compounded 30 times. Far-away cash flows are also very hard to forecast.

Check yourself – Part 2

- Q4. Project: cost 500 now, then receives 200, 250, 300 at the ends of years 1, 2, 3. Rate 10%. Compute NPV. Accept?
- Q5. A firm will earn 2 million next year, growing 2% yearly, cost of capital 10%. Rough value?
- Q6. Why does a person who wants cash today still prefer to take a positive-NPV project?

Part 3. Other capital budgeting rules (Lecture 3B)

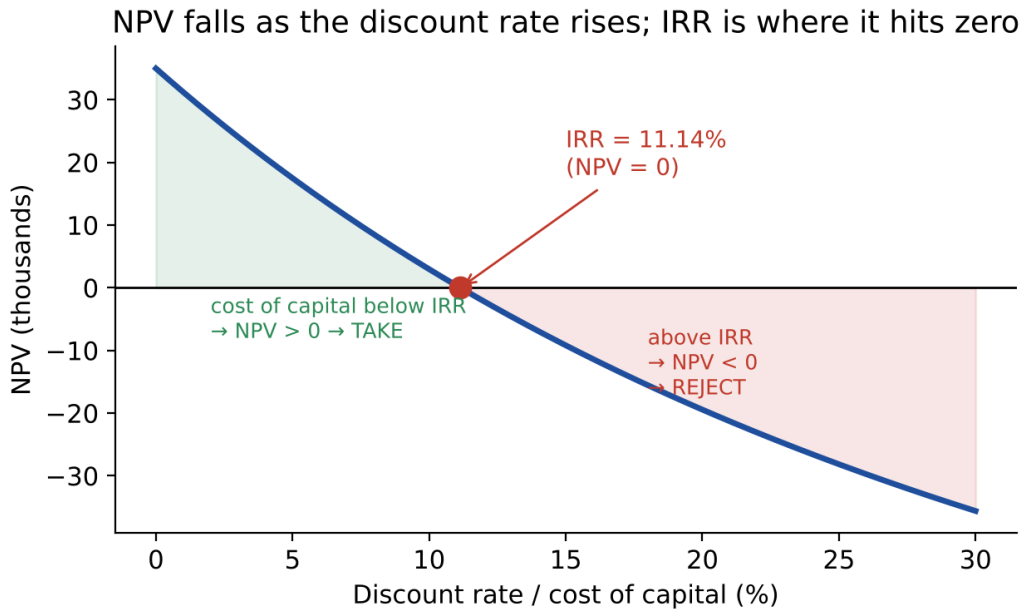
Capital budgeting = deciding which projects to invest in. NPV is one rule; these are three more that managers use. You need to know what each does well and where it fails.

3.1 Internal Rate of Return (IRR)

Big idea
IRR is the project's own rate of return: the interest rate that makes NPV exactly zero. **Rule:** if IRR > cost of capital (called the **hurdle rate**), take the project.

Worked example: One outflow, one inflow
Cost 100, pays 120 next year. Solve $-100 + 120 \div (1+\text{IRR}) = 0 \rightarrow$ IRR = **20%** (just the simple return). If the market rate is 10%, 20% > 10% → take it. NPV agrees.

Worked example: Several cash flows
Cost 100,000; receives 5,000 (yr 1), 10,000 (yr 2), 120,000 (yr 3). Solve $-100{,}000 + 5{,}000 \div (1+\text{IRR}) + 10{,}000 \div (1+\text{IRR})^2 + 120{,}000 \div (1+\text{IRR})^3 = 0$.
Answer: **IRR = 11.14%** (by trial and error or Excel's IRR function).
If the cost of capital is 11.25% → reject. If 11% → accept. NPV gives the same verdicts.



NPV of the 100,000 project vs discount rate. Where the curve crosses zero is the IRR.

Advantages of IRR

- It depends only on the project's cash flows, so you can compute it without looking at markets or interest rates.
- It helps judge project profitability and a manager's performance.
- Comparing IRR with the cost of capital shows your **buffer**: how wrong your cash flow estimates can be before the decision flips.

Where IRR goes wrong

Problem	Example from lecture	What to do
Multiple IRRs (cash flows switch sign more than once)	Cost 100,000 now; +205,000 in yr 1; environmental clean-up −102,000 in yr 2. Both −15% and 20% give NPV = 0.	Spreadsheets may show only one. Graph the NPV instead.
No IRR at all	Receive 100 today and 200 next period (no outflow), or: −10,000, +27,000, −20,000. NPV is never zero.	Spreadsheet gives an error. Use NPV.
Financing, not investing	You *receive* 100,000 now and pay 5,000, 10,000, 120,000 later. IRR is still 11.14%.	Flip the rule: take it only if IRR is **below** the market rate (cheap loan).
Mutually exclusive projects	Project 1: invest 5, get 10 (IRR 100%). Project 2: invest 1,000, get 1,100 (IRR 10%). Rate 5%.	NPV of 1 = 4.52, NPV of 2 = 47.62. **Project 2 is better** despite the lower IRR. IRR ignores scale.
Different rates for different horizons	Short-term and long-term costs of capital differ.	IRR has no single rate to compare with. Use NPV.

Safe zone
IRR is safest when there is just **one outflow followed only by inflows** (one sign change). Then there is one and only one IRR.

3.2 Profitability Index (PI)

Formula
PI = (present value of FUTURE cash flows) ÷ (project cost) PI > 1 ⇔ NPV > 0

It measures **value per unit of money invested**. It only makes sense if the first cash flow is an outflow. Interest rate 20% in all three:

Project	Cash flows (yr 0,1,2,3)	PV of future flows	PI	NPV
1	−100, 70, 60, 50	128.94	1.29	28.94
2	−10, 21.14, 18.12, 15.10	38.94	3.89	28.94
3	−100, 140, 120, 100	257.87	2.58	157.87

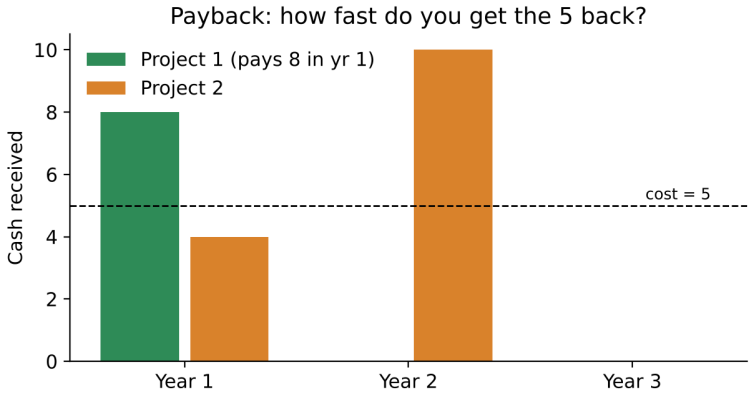
- Projects 1 and 2 have the **same NPV** (28.94), but Project 2 needs only 10 upfront, so its PI is much higher. Prefer Project 2.
- Project 3 has a lower PI than Project 2 but a far higher NPV (157.87). So PI is **biased towards small projects**.
- **Use PI to choose between projects only when their NPVs are equal or very similar.**

3.3 Payback rule

Payback period = how many years until you get your original investment back. Shorter is "better".

Project	Cash flows	Payback	NPV @5%
1	−5, +8 in yr 1	1 year	lowest

Project	Cash flows	Payback	NPV @5%
2	−5, +4 in yr 1, +10 in yr 2	2 years	middle
3	−5, +4, 0, then +1,000,000 in yr 3	3 years	**highest**



Projects 1 and 2 (cost 5). Fast payback does not mean the biggest value.

Weakness: it ignores all cash flows after payback (Project 3's million) and ignores the time value of money. **Strengths:** easy to explain; useful as a control when managers aren't fully trusted; helpful for an entrepreneur with limited capital facing an imperfect market (money is expensive, so getting it back soon matters). **Verdict: supplementary information, not the main decision rule.**

3.4 Quick comparison

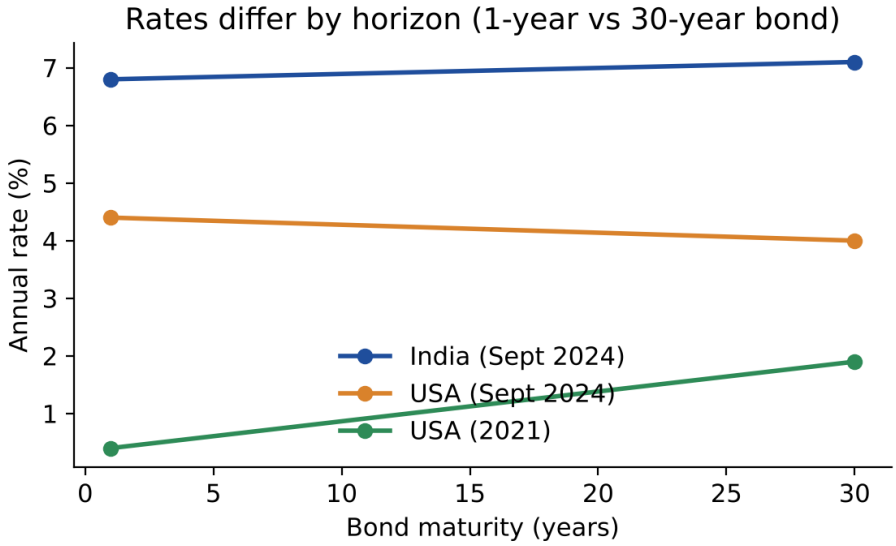
Rule	Question it answers	Main weakness
NPV	How much value (in today's money) does it add?	Needs good estimates of cash flows and discount rate
IRR	What is the project's own rate of return?	Multiple/no IRRs; misleading for exclusive projects
Profitability Index	Value created per unit invested	Biased to small projects
Payback	How fast do I get my money back?	Ignores later cash flows and time value

Check yourself – Part 3

- Q7. Cost of capital is 10%. Project A: −100, +60, +60. Project B: −100, +10, +20, +200. Find NPV and PI of each, and payback. Which would you pick and why?
- Q8. Using Q4's cash flows (−500, 200, 250, 300), find the IRR (use a calculator/Excel). Is it above 10%?
- Q9. True or false: a project with a 100% IRR always beats one with a 10% IRR. Explain.

Part 4. Rates that change over time (Lecture 3C)

So far we assumed one interest rate for all horizons. In reality a 1-year and a 30-year bond pay different yearly rates, and the rate next year can differ from this year's.



Rates quoted in the lecture: India and US government bonds (Sept 2024) and the US in 2021.

4.1 Compounding different yearly rates

Formula
$1 + r_{0,2} = (1 + r_{0,1}) \times (1 + r_{1,2})$ ($r_{a,b}$ = return from year a to year b)

- **Spot rate**: an interest rate that starts *now*. **Forward rate**: an interest rate that starts in the *future*.

Worked example: Two years
30% in year 1, 20% in year 2: (1.30)(1.20) = 1.56 → two-year holding return **56%** (not 50%).

Worked example: Three years starting next year
Yearly returns 30%, 40%, 50%: (1.3)(1.4)(1.5) = 2.73 → holding return **173%**. So 100 invested in year 1 becomes 273 in year 4.

4.2 Annualized rate of return

To compare "10% over 3 years" with "20% over 6 years" you need a **per-year** figure. The annualized rate is the single yearly rate that, compounded, gives the same total.

Formula
$1 + \text{annualized rate} = (1 + \text{total holding return})^{1/t}$ (the t-th root)

Worked example: 173% over 3 years
$(2.73)^{1/3} = 1.3976$ → **39.76% per year**. The naive answer 173% ÷ 3 = 57.7% is wrong: $1.577^3 - 1 =$ 292%, because it ignores interest on interest.

Worked example: 100 becomes 240 in 30 years
Total return 140%. $2.4^{1/30} = 1.0296$ → **2.96% per year**.

Memory trick
Compounding ≈ **adding** returns. Annualizing ≈ **averaging** returns. (Earning 1% twice compounds to 2.01% ≈ 1% + 1%; annualized back is 1%.)

4.3 Why plain averages mislead



Average of +20% and −20% = 0% BUT you ended with a loss of 4%
(annualized return = −2.02% per year)

A +20% year followed by a −20% year averages 0%, but you lose money.

- **Doubling then halving**: +100% then −50%. The average is +25%, but you end where you started: true return 0%.
- **+20% then −20%**: 1000 → 1200 → 960. Average says 0%; annualized rate = $\sqrt{(0.96)} - 1$ = **−2.02%**, correctly showing a loss.
- So always judge multi-year performance by compounded (annualized) returns, not simple averages.

4.4 NPV with time-varying rates

Formula
$NPV = C_0 + C_1 \div (1+r_{0,1}) + C_2 \div (1+r_{0,2}) + \ldots$ Each cash flow gets its **own** discount rate for its own date.

Worked example: Project with different rates
Cost 24 now; pays 20 in 1 year and 16 in 5 years. 1-year rate 5%; 5-year annualized rate 6%.
$NPV = -24 + 20 \div 1.05 + 16 \div 1.06^5 = -24 + 19.05 + 11.96 \approx$ **7.01** → take it.

Check yourself – Part 4

- Q10. Yearly returns of 10%, 20% and 5%. Find the 3-year holding return and the annualized rate.
- Q11. An investment grows 50% over 2 years. What is the annualized return?
- Q12. An investment gains 50% and then loses 50%. What is the average return, and what actually happens to 100 invested? Annualized return?
- Q13. Cost 50 now; pays 30 in 1 year and 40 in 3 years. The 1-year rate is 4% and the 3-year annualized rate is 5%. NPV?

Part 5. Formula cheat sheet

Topic	Formula	Remember
Future value	$FV = C(1+r)^t$	Grow forward
Present value	$PV = C \div (1+r)^t$	Shrink back to today
Annuity (loan)	$PV = (C \div r)[1 - (1+r)^{-n}]$	r and n must use the same period (months!)
Quoted annual rate (banks)	monthly = quoted ÷ 12	No compounding in the quote
Interest in a payment	rate × current balance	Rest is principal
Constant perpetuity	$PV = C \div r$	
Growing perpetuity / Gordon	$PV = C_1 \div (r - g)$	Need r > g
Dividend yield	$D \div P = r - g$	Infer r from the market
NPV	$\Sigma\, C_t \div (1+r)^t$	Take if > 0
IRR	rate where NPV = 0	Take if IRR > cost of capital (for investments)
Profitability Index	PV(future flows) ÷ cost	> 1 ⇔ NPV > 0; biased to small projects
Payback	years to recover cost	Ignores later flows
Holding return (two periods)	$(1+r_{0,1})(1+r_{1,2}) - 1$	Multiply, don't add
Annualized return	$(1+r_{0,t})^{1/t} - 1$	t-th root

Decision flow

- 1. List all cash flows and dates (outflows negative).
- 2. Pick the right discount rate for each date.
- 3. Compute NPV. Positive → accept; negative → reject.
- 4. Use IRR, PI and payback only as extra information (sanity checks, buffers, small-capital situations).
- 5. Remember inputs are estimates: run a sensitivity check, especially for long-term projects.

Part 6. Practice problems

Try each without peeking. Use a calculator. Round money to 2 decimals and rates to 2 decimals.

Q1 (PV). What is 500 received in 2 years worth today at a 10% yearly rate?

Q2 (Loan). You borrow 200,000 for 20 years with equal monthly payments at a quoted 9% per year. (a) Find the monthly payment. (b) How much of the first payment is interest and how much is principal?

Q3 (Gordon). A stock will pay a dividend of 50 next year. Dividends grow 4% forever and investors require 9%. What is a fair price?

Q4 (NPV). A project costs 500 today and pays 200, 250 and 300 at the end of years 1, 2 and 3. Discount rate 10%. Compute NPV. Should you accept?

Q5 (Perpetuity). A firm expects profits of 2 million next year, growing at 2% per year. Cost of capital is 10%. Estimate its value.

Q6 (Concept). In a perfect capital market, why should someone who wants money today still accept a positive-NPV project that pays off next year?

Q7 (Rules). Cost of capital 10%. Project A: −100, +60, +60. Project B: −100, +10, +20, +200. Find NPV, PI and payback for each. Which do you choose?

Q8 (IRR). Find the IRR of −500, +200, +250, +300 (trial and error or Excel). Compare it with 10%.

Q9 (IRR trap). True or false: you should always pick the project with the higher IRR. Give a short reason or a counter-example.

Q10 (Compounding). Yearly returns are 10%, 20%, 5%. Find the 3-year holding return and the annualized return.

Q11 (Annualizing). An investment returns 50% in total over 2 years. What is the annualized return?

Q12 (Averages). An investment gains 50% then loses 50%. (a) Average return? (b) What happens to 100? (c) Annualized return?

Q13 (Varying rates). Cost 50 now; pays 30 in 1 year and 40 in 3 years. 1-year rate 4%; 3-year annualized rate 5%. Find NPV.

Q14 (Sensitivity). A project pays 100 in 1 year (rate 5%). Another pays 100 in 30 years (rate 5%). If you mistakenly use 5.5% in both, which present value is affected more, in percentage terms? (Compute both.)

Part 7. Answer key

Q1. PV = 500 ÷ 1.10^2 = 500 ÷ 1.21 = **413.22**.

Q2. Monthly rate = 9% ÷ 12 = 0.75%, n = 240. Payment = 200,000 × 0.0075 ÷ [1 − 1.0075^{-240}] = **1,799.45**. First-month interest = 0.75% × 200,000 = **1,500.00**; principal = 1,799.45 − 1,500 = **299.45**.

Q3. P = 50 ÷ (9% − 4%) = 50 ÷ 0.05 = **1,000**.

Q4. NPV = −500 + 200÷1.1 + 250÷1.21 + 300÷1.331 = −500 + 181.82 + 206.61 + 225.39 = **113.82**. Positive → accept.

Q5. Value ≈ 2,000,000 ÷ (10% − 2%) = **25,000,000** (rough estimate).

Q6. Take the project, then sell it or borrow against it in the market. You receive its present value today, which exceeds the cost, so you have more cash today than if you skipped it. Timing preferences are handled by the market (separation of decisions).

Q7. A: NPV = −100 + 54.55 + 49.59 = **4.13**; PI = 104.13 ÷ 100 = **1.04**; payback = 2 years (1.67 if cash arrives evenly). B: NPV = −100 + 9.09 + 16.53 + 150.26 = **75.88**; PI = **1.76**; payback = 3 years. **Choose B**: far higher NPV even though A pays back sooner.

Q8. IRR ≈ **21.65%**. It is above 10%, so accept (agrees with NPV in Q4).

Q9. False. IRR ignores project size. Lecture example at 5%: invest 5 get 10 (IRR 100%, NPV 4.52) vs invest 1,000 get 1,100 (IRR 10%, NPV 47.62). If only one can be taken, take the higher NPV.

Q10. (1.10)(1.20)(1.05) = 1.386 → holding return **38.6%**. Annualized = $1.386^{1/3}$ − 1 ≈ **11.49%** (not 38.6 ÷ 3 = 12.87%).

Q11. $1.5^{1/2}$ − 1 = **22.47%** per year (not 25%).

Q12. (a) Average = (50% + (−50%)) ÷ 2 = 0%. (b) 100 → 150 → 75, a **25% loss**. (c) Annualized = √0.75 − 1 = **−13.40%** per year.

Q13. NPV = −50 + 30 ÷ 1.04 + 40 ÷ 1.05^3 = −50 + 28.85 + 34.55 = **13.40** → accept.

Q14. 1-year: correct PV = 95.24, wrong PV = 94.79 → error ≈ **0.47%**. 30-year: correct = 23.14, wrong = 20.06 → error ≈ **13.3%**. The long-term project is far more sensitive to the rate.

Source: your lecture slides FF_3A, FF_3B and FF_3C (Corporate Finance, based on Chapters 2–5 of Ivo Welch's textbook). Worked examples use the lecture numbers; practice problems are new.